An evaluated testing dictionary for the positive-angular C1 main Gram

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This calculation concerns the main Gram of the all-H, bounded-z C1 certificate at public commit `0c1ce2e15ed067a6845464e2468559cea5274f2c`. It gives fixed rational bounds on a genuine source/testing rectangle, including source points approaching $D = 0$, $\beta = 0$, and $H = +\infty$. The condition $\lambda > 0$ is retained. The separate $\lambda \le 0$ construction and the retained squares outside this Gram are not compiled here. Thus this is not yet a local whole-parent pencil or a full-set pencil.

1 The rectangle and source formulas

Fix the rational constants

$$\varepsilon = 2^{-160}, \qquad \delta = 2^{-100}, \qquad c_0 = 2^{-8}, \qquad A_* = 2^{30}, \qquad \gamma = \frac{c_0}{14A_*} = \frac{1}{14\,2^{38}}.$$

The source domain consists of

$$0 < x, \sigma \le \varepsilon, \quad 0 < z \le 1, \quad 0 < b < 1, \quad r = bz < 1, \quad 0 < h < 1,$$

with the original strict gaps and the two nonnegative event coefficients specified below. Set

$$Q = 1 - b, \quad \theta = 1 - h, \quad H = h/\sigma, \quad e = xh/\sigma, \quad E = 1 - e > 0, \quad k = \sqrt{x}, \quad q = \sqrt{z},$$

$$K = 1 + x, \quad R_s = 1 + xr, \quad M = R_s - xz\theta > 0, \quad J = 1 + xh - \sigma(1 - r) > 0,$$

$$P = \sigma + K + \sigma z(1 + bx + 2xh) + \sigma b x^2 h z^2 + x^2 h\theta z,$$

$$p = \frac{z(1 + xh)E + (1 - r)(1 + xhz)}{2P}, \qquad l = \frac{(1 + \sigma r)\{eR_s(1 + \sigma z) + ME\}}{2P},$$

$$\kappa = \frac{K(1 + \sigma r)(1 + \sigma z)}{2MP}.$$

The original coefficients are

$$m = \kappa x^2 z H = \kappa x z e, \qquad C = \frac{Kz(1 + \sigma r)E}{2(1 - r)P},$$

$$A = p/r, \quad \alpha = l/(zQ), \quad a = p/b, \quad \ell = l/Q,$$

$$D = \frac{K(1 + \sigma z)(1 + \sigma r) - JP}{2JP} \ge 0, \qquad \beta = \frac{K(1 + \sigma r)(1 + xhz) - \theta P}{2\theta P} \ge 0.$$

Here H has no upper bound. Neither b, Q, θ, $1 - r$, nor z is bounded below. The testing rectangle is independently

$$0 \le u, w \le \delta.$$

For signed angles u_s, w_s, set $\alpha_s = \operatorname{sgn}(u_s)$ and $\beta_s = \operatorname{sgn}(w_s)$, choosing the value 1 at zero. The exact testing gauge is

$$(u, w, X, Y, Z, W) = (|u_s|, |w_s|, \beta_s X_s, \alpha_s Y_s, \alpha_s\beta_s Z_s, W_s).$$

It preserves the parent polynomial. Composing every testing function below with this one fixed piecewise semialgebraic map extends the same feature count to $|u_s|, |w_s| \le \delta$; no square root of a negative angle is taken. State coordinates X, Y, Z, W are unrestricted real numbers; no state ratio is bounded.

Define

$$H_0 = 1 + zQ, \quad B_s = P + xz(J - \sigma),$$
$$\mathcal{H} = R_s H_0 + z\{(1-r)xh(2+xr) - e[2 + x(1 + h + r\theta)] + Ke^2\}.$$

Two exact source identities, obtained by multiplication, are

$$2\theta P\beta = hB_s - \sigma H_0, \tag{1}$$
$$2P(JD + x\theta\beta) = \sigma^2\mathcal{H}. \tag{2}$$

The angular quantities used below are

$$\nu = (2D - 2\beta\theta + h)/\sigma, \quad \lambda = (8Dh + 4\beta\theta^2 + 2h^2)/\sigma - 4bp\sigma z > 0,$$
$$K_A = 4\kappa H - \lambda, \quad d_G = 4D/\sigma^2, \quad \rho = \lambda x/\sigma, \quad S = d_G + \rho,$$
$$L_0 = \lambda\sigma, \quad q_0 = \lambda k, \quad c = \max(0, 1 - K_A), \quad a_0 = k/\sigma.$$

The letter q_0 here is a source quantity, not the constant coefficient of a later scalar channel.

The complete reciprocal parent being certified is

$$\begin{aligned} F_w = {} & 4m(W - wX + uY - uwZ)^2 + 4C(1 + w^2)(uW - Y)^2 \\ & + 4A(1 + u^2)(1 + w^2)Y^2 + (4D + 2)(uwW + uX - wY - Z)^2 \\ & + (4\alpha + 2)(1 + u^2)(Y - wZ)^2 + (4\beta + 2)(1 + w^2)(X + uZ)^2 \\ & - 2(1 + u^2)(1 + w^2)Z^2. \end{aligned}$$

All four state coordinates occur; they need not be normalized.

2 Uniform source bounds, without an upper bound on H

All inequalities in this section follow under the larger condition $\varepsilon \le 2^{-22}$, so in particular on the chosen rectangle. Directly from the positive formula for P and the definitions,

$$1 \le P \le 1 + 5\varepsilon, \quad |J - 1|, |M - 1| \le \varepsilon, \quad 1 \le B_s \le 1 + 7\varepsilon, \quad 1 \le H_0 \le 2.$$

Initially $0 < e < 1$ gives $0 \le \mathcal{H} \le 4$: nonnegativity uses (2). Eliminate β to obtain

$$eB_s = xH_0 + \sigma\mathcal{H} - 2PJD/\sigma.$$

Consequently

$$e \le 2x + 4\sigma \le 6\varepsilon, \qquad |\mathcal{H} - H_0| \le 32\varepsilon, \qquad 0 \le D \le 4\sigma^2, \quad 0 \le \beta\theta \le 1. \tag{3}$$

For example, in $\mathcal{H} - H_0$ the terms are bounded by $2\varepsilon + 3\varepsilon + 3e + 2e^2$, which is at most 32ε here. The same positive formulas give

$$|p - H_0/2| \le 16\varepsilon, \quad |l - 1/2| \le 8\varepsilon, \quad 1/4 \le p, l, \kappa \le 2, \quad \kappa \le 1, \quad m \le 6\varepsilon^2, \quad C(1 - r) \le 1. \tag{4}$$

In particular $4a, 4\ell \geq 1$. The unbounded positive coefficients a, ℓ, C will be retained in whole squares and will not be estimated separately.

Exact rearrangement gives

$$\nu = 2D/\sigma + [H_0 - ez(J - \sigma)]/P,$$

$$\lambda = 2H + 2e\theta z(J - \sigma)/P - 2\theta H_0/P + 8Dh/\sigma - 4bp\sigma z.$$

Thus

$$|\nu| \leq 3, \qquad |\lambda - 2H| \leq 5, \qquad |L_0 - 2h| \leq 8\sigma. \tag{5}$$

For completeness, the cancellation needed when H is large is polynomial. Write

$$N_\kappa = K(1 + \sigma r)(1 + \sigma z) - MP = xF_x + \sigma F_\sigma.$$

Here $F_\sigma = r - 1 + \sigma rz$, and F_x is the polynomial $(N_\kappa - N_\kappa|_{x=0})/x$. Its coefficient ℓ^1 norm in (x, σ, z, b, h, r) is 37, as can also be checked by expanding the displayed product. Since all six arguments lie in $[0, 1]$,

$$H(2\kappa - 1) = \frac{eF_x + hF_\sigma}{MP}, \qquad |H(2\kappa - 1) - h(r - 1)| \leq 462\varepsilon.$$

The estimate $|MP - 1| \leq 8\varepsilon$ suffices for the second inequality. Combining this with (5) more precisely yields

$$|K_A - \{2\theta H_0 - 2h(1 - r)\}| \leq 2^{12}\varepsilon, \qquad |K_A| \leq 5, \quad 0 \leq c \leq 6. \tag{6}$$

If $c > 0$, then $h \geq 1/8$: for $h < 1/8$ the expression in braces is at least $3/2$, whereas the error is less than $1/4$. The earlier elimination identity also gives

$$a_0^2(hB_s - \sigma H_0) \leq \mathcal{H}.$$

Therefore, whenever $c > 0$,

$$a_0 \leq 8, \qquad L_0 \geq 1/8, \qquad 1/\lambda \leq 8\sigma. \tag{7}$$

Every term below containing a_0 or $1/\lambda$ from the angular subtraction is multiplied by c. On $c = 0$ that subtraction is identically zero; no uniform estimate on those unused factors is required.

Put

$$R_0 = -4\{2(C + D)(r - 1) + 2p + r - 1\}.$$

The positive expression for $C(1 - r)$ and the formula for p give $|R_0| \leq 256(x + \sigma) \leq 512\varepsilon$. Since

$$(x + \sigma)^2 H = xe + 2\sigma e + \sigma h \leq 2\varepsilon,$$

(5) further gives

$$|R_0|\sqrt{\lambda} \leq 1024\sqrt{\varepsilon}. \tag{8}$$

Finally the exact identity

$$S = \frac{2\mathcal{H}}{P} + d_G(1 - J + 2hx) + \frac{2eH_0}{P} - \frac{2e^2 z(J - \sigma)}{P} - 4bpxz$$

shows

$$|S - 2H_0| \leq 256\varepsilon, \qquad 1 \leq S \leq 5, \quad 0 \leq d_G, \rho \leq 5, \quad 0 < L_0 \leq 3, \quad 0 < q_0 \leq 4. \tag{9}$$

Here $q_0^2 = L_0\rho$; all bounds use $\lambda > 0$ but no lower bound on λ or upper bound on H.

3 A finite formula for the main Gram

The state roots are

$$N_x = X + uhZ, \quad N_y = (Y - bwZ)/q, \quad R = wZ/q - uqW, \quad \Gamma = \sigma Z - uwW,$$
$$A_\theta = q(u^2 - x)W, \quad T_\theta = kq(u - k)W, \quad J_C = N_y + bR, \quad J_\ell = N_y - QR, \quad \Gamma_* = \sqrt{d_G}\Gamma.$$

Let

$$B_1 = \sqrt{\lambda\sigma}\, uZ, \quad B_2 = \sqrt{\rho}\, uwW, \quad B_3 = \sqrt{\lambda}\, xqW, \quad B_4 = \sqrt{\lambda x}\, wZ/q.$$

When $WZ > 0$, put $G_* = \operatorname{sgn}(W)(B_1B_2B_3B_4)^{1/4}$ and $d_i = B_i - G_*$. This is the positive-state convention followed, when necessary, by changing the signs of all roots together. When $WZ \le 0$ put $d_i = B_i$. Equivalently, for every state,

$$G_* = \sqrt{\lambda x}\, \operatorname{sgn}(W)\sqrt{uw\max(WZ, 0)}, \qquad d_i = B_i - G_*.$$

Define

$$U_* = \sqrt{q_0}\sqrt{u}\, q(u - k)W, \qquad K_* = \sqrt{q_0}\sqrt{k}\, q(u - k)W.$$

Use the twelve-root column

$$g = (N_x, J_C, J_\ell, \Gamma_*, A_\theta, T_\theta, d_1, d_2, d_3, d_4, U_*, K_*)^{\mathsf{T}}.$$

The two radial changes are source-only and uniformly bounded:

$$R = J_C - J_\ell, \qquad N_y = QJ_C + bJ_\ell.$$

Set

$$D_* = \operatorname{diag}(2, 1, 1, 1, 1/2, 1/2, 1/2, 1/2, 1/2, 1/2, 1, 1/2).$$

We now define a polynomial quadratic form $\mathcal{E}(g)$, which fully specifies the correction matrix; a cross term is divided by two when placed in its two symmetric entries.

For its coefficients put $t = u^2$, $\xi = w^2$, $g_s = h - Q$ and

$$\begin{aligned}
R_x &= -4\{2Db + (1 + 2m)g_s - 2\beta\theta\}, \quad B_x = 8\kappa e,\\
A_y &= 8(p - l) + 4z\{(1 + 2m)g_s + 2Dh + 2p\sigma\},\\
\chi &= 8(pb + lQ) - 4h + 4eh(2D + 1) - 8Dh\\
&\quad + 4z\{4Dbh + 4pb\sigma + (1 + 2m)g_s^2 + 2\beta\theta^2\},\\
k_1 &= 2(1 + 2m)g_s^2 + 8Dhb + 8pb\sigma + 4\beta\theta^2.
\end{aligned}$$

Then

$$\begin{aligned}
E_{yy} &= 2z\{(1 + 2m)u^2 + (1 + 2D)w^2\},\\
E_{RR} &= z\{8bD + 8bp\sigma + 4b\sigma(2C + 2D + 1)(r - 1) + k_1u^2 + 2b^2(2D + 1)w^2\},\\
E_{yR} &= 4z\{u^2[2Dh + 2p\sigma + (1 + 2m)g_s] + b(2D + 1)w^2\\
&\qquad + \sigma(2C + 1)(r - 1) + 2D[1 + \sigma(r - 1)] + 2p\sigma\}.
\end{aligned}$$

The entire first normal is represented by the linear form

$$F_1 = \frac{\sqrt{d_G}}{S}u\Gamma_* + \frac{\sqrt{\rho}}{S}\{kd_1 - ud_2 + (u - k)d_3\} - \frac{\rho}{S}\sqrt{\sigma}T_\theta.$$

The entire second normal is represented by

$$F_2 = \frac{R_0\sqrt{d_G}}{S} qw\Gamma_* + \frac{R_0\rho\sigma z}{S} R - \frac{R_0\sqrt{\rho}}{S}(qw + z\sqrt{\sigma})(d_2 - d_3) + \left\{ \frac{R_0 q_0 z}{S} + \frac{R_0\sqrt{\lambda}\sqrt{\rho}}{S} qw \right\} T_\theta - 8pquwF_1.$$

These are exact identities $F_1 = u\Gamma$ and $F_2 = qw(R_0 - 8pu^2)\Gamma$. In particular they keep both complete state-normal couplings, not only their contact restrictions.

Write

$$\varepsilon_C = 2/\lambda + u + k, \qquad \varepsilon_G = 2a_0 u + (u + k)/L_0.$$

The correction is exactly

$$\begin{aligned}\mathcal{E} = {} & E_{yy}N_y^2 + E_{RR}R^2 + E_{yR}N_yR - 4\nu N_xF_1 + J_CF_2 \\ & - 4uqw(2m + 2D + 1)N_xN_y + uqwR_xN_xR + qwB_xN_xA_\theta + uA_yN_yA_\theta + \chi uA_\theta R \\ & - \tfrac{1}{2}\rho\sigma R^2 + \sqrt{q_0}\sqrt{u}\,U_*R - \frac{cd_G}{S}\varepsilon_C \sum_{i=1}^{4} d_i^2 - \frac{cd_G}{S}u^2R^2 - \frac{c}{S}\varepsilon_G\Gamma_*^2 \\ & - \frac{c}{S}\{\sqrt{a_0u}\,U_* + \sqrt{a_0k}\,K_*\}^2.\end{aligned}$$

Define G by $g^\mathsf{T}Gg = g^\mathsf{T}D_*g + \mathcal{E}$. This is the actual main Gram after retaining the positive excess of the N_x, J_C, J_ℓ squares, splitting $K_A + c \geq 1$ equally between A_θ, T_θ, and retaining their nonnegative difference. The averaged circuit identity and angular identity used in this step are precisely those of the printed all-H proof at lines 7550–7740. This definition therefore prints every main-Gram entry through one quadratic polynomial instead of twelve rows of expanded rational functions.

4 Rational coefficient bounds and fixed radicals

Put $v = \sqrt{u}$. Every entry of $G - D_*$ is a polynomial in (v, w), with source coefficients, supported on

$$\{1, w, w^2, v, v^2, v^2w, v^4, v^4w\} \subset \{v^\alpha w^\beta : 0 \leq \alpha \leq 4, 0 \leq \beta \leq 2\}.$$

The following elementary bounds provide a reproducible coefficient estimate:

$$|R_x|, |A_y| \leq 32, \quad |\chi| \leq 64, \quad |B_x| \leq 48\varepsilon, \quad |k_1| \leq 16.$$

The coefficients of E_{yy} on (u^2, w^2) are bounded by $(4, 4)$; those of E_{RR} on $(1, u^2, w^2)$ by $(64\sigma, 16, 4)$; those of E_{yR} by $(32\sigma, 12, 8)$. These follow term by term from (3)–(4); in the constant terms use $C(r - 1)$, never C separately.

Here is a single positive-coefficient majorant which verifies all remaining estimates. In it $s = \sqrt{\varepsilon}$, all twelve root letters are formal nonnegative indeterminates, and $n = r' = J_C + J_\ell$. Define

$$f_1 = u\Gamma_* + sd_1 + ud_2 + (u + s)d_3 + sT_\theta,$$

$$f_2 = 512s^2w\Gamma_* + 512s^4r' + 512s^2(w + s)(d_2 + d_3) + (2048s^2 + 1024sw)T_\theta + 16uwf_1.$$

The absolute coefficient polynomial of $\mathcal{E}$ is bounded by

$$\mathcal{Q} = (4u^2 + 4w^2)n^2 + (64s^2 + 16u^2 + 4w^2)r'^2 + (32s^2 + 12u^2 + 8w^2)nr'$$

$$+ 12N_x f_1 + J_C f_2 + 16uwN_x n + 32uwN_x r' + 48s^2 wN_x A_\theta + 32unA_\theta + 64uA_\theta r'$$
$$+ \tfrac{5}{2}s^2 r'^2 + 2vU_* r' + 6(16s^2 + u + s)\sum_i d_i^2 + 6u^2 r'^2 + (144u + 48s)\Gamma_*^2$$
$$+ 48uU_*^2 + 96vU_* K_* + 48sK_*^2.$$

Indeed (9) bounds $\sqrt{d_G}/S$, $\sqrt{\rho}/S$, d_G/S, ρ/S by one, while (7) gives $\varepsilon_C \le 16\sigma + u + k$ and $\varepsilon_G \le 24u + 8k$ when needed. The two small coefficients in F_2 use $|R_0 q_0| \le 2048\varepsilon$ and (8).

Expansion, or simply evaluation at one of this positive-coefficient polynomial, gives coefficient ℓ^1 norm 8590. Its constant testing part has norm 5030 and every term of that part is divisible by s. Consequently every source coefficient of every nonconstant testing monomial in G has absolute value below 2^{20}, and the source matrix $G(0,0)$ satisfies

$$|G_{ij}(0,0) - (D_*)_{ij}| \le 2^{20}\sqrt{\varepsilon} = 2^{-60}. \tag{10}$$

These estimates hold before any ordering of the tensor product is assumed. They bound pure-source coefficients in their own field.

For $r = 12$ the following exact quadratic-form decomposition uses only fixed directions:

$$G = \sum_i \left(G_{ii} - \frac{11}{48}\right) e_i e_i^{\mathsf{T}} + \sum_{i<j} \frac{1/48 + G_{ij}}{2}(e_i + e_j)(e_i + e_j)^{\mathsf{T}} + \sum_{i<j} \frac{1/48 - G_{ij}}{2}(e_i - e_j)(e_i - e_j)^{\mathsf{T}}.$$

Write each of its 144 scalar channels as

$$q(c,t) = q^{(0)}(c) + \sum_{j=1}^{14} a_j(c) b_j(t),$$

where b_j run through the nonconstant monomials in the 5×3 rectangle above. By (10), for every channel

$$q^{(0)} \ge c_0 = 1/256, \qquad |a_j| \le A_* = 2^{30}.$$

In fact the off-diagonal channels have constant term at least $1/96 - 2^{-61}$ and the diagonal ones at least $13/48 - 2^{-60}$. On the independent testing rectangle, $0 \le b_j \le \sqrt{\delta} = 2^{-50} < \gamma$. Hence the exact separated identity

$$q = (q^{(0)} - c_0) + \frac{1}{2}\sum_{j=1}^{14}\big[(A_* + a_j)(\gamma + b_j) + (A_* - a_j)(\gamma - b_j)\big]$$

has nonnegative factors in their own labelled fields. It proves the required tensor SOS without an appeal to entrywise positivity in a tensor ring or to a point-dependent real congruence.

All twelve roots are source-linear combinations of the fixed testing functions

$$\mathcal{V} = \{X, Y, Z, W, uZ, wZ, uW, uwW, u^2 W, u^{3/2} W, u^{1/2} W, \operatorname{sgn}(W)\sqrt{uw\max(WZ, 0)}\}.$$

The last function includes both state-sign cases and is zero when $WZ \le 0$. Changing all state signs changes all twelve roots' signs and preserves the quadratic form. Therefore a fixed main-Gram dictionary is

$$\mathcal{V} \;\cup\; \{f\sqrt{\gamma + b_j}, f\sqrt{\gamma - b_j} : f \in \mathcal{V},\ 1 \le j \le 14\}.$$

It has 348 displayed functions. All radicals are testing-only; all remaining coefficients are source-only square roots.

Only seven nonconstant monomials actually occur. Dropping the zero channels and replacing γ by $1/(7\,2^{38})$ gives the smaller displayed list of $12(1+14) = 180$ functions. This optional reduction changes no source estimate. The accompanying exact polynomial check records the support as well as both norms.

The separate multiplier of the retained circuit remainder is $\frac{1}{2} - cd_G\varepsilon_C/S$. Its constant term is at least $1/2 - 6(16\varepsilon + \sqrt{\varepsilon}) > c_0$ and its only testing coefficient is bounded by six. More explicitly, put

$$a_C = -cd_G/S, \qquad q_C^{(0)} = \tfrac{1}{2} - (cd_G/S)(2/\lambda + k).$$

Then its fixed-radical factorization is

$$q_C^{(0)} + a_C u = (q_C^{(0)} - c_0) + 13A_*\gamma + \tfrac{1}{2}\{(A_* + a_C)(\gamma + u) + (A_* - a_C)(\gamma - u)\}.$$

Every displayed pure factor is nonnegative on the rectangle. Its circuit remainder roots, and the other retained squares, still require their own testing features; they are not counted in 348 or 180.

5 The two zero faces are present, and H is unrestricted

No inverse of D, β, b, Q or a state coordinate was used in the coefficient estimates or Gram decomposition. At $D = 0$ the root Γ_* is zero and all formulas remain literal. At $\beta = 0$ the same formulas apply. Testing axes are included. The denominators S, M, P, J, σ, k used above are nonzero on the stated source domain; their potentially dangerous combinations were bounded explicitly rather than omitted on a boundary.

For explicit nonemptiness set $x = \sigma = t$, $z = 1$, $b = 1/2$, $h = Ht$, with $0 < t \le \varepsilon$. The numerators $D_N = 2JPD$ and $\beta_N = 2\theta P\beta$ are exactly

$$\begin{aligned} D_N &= (3 - H)t^2 + (3/4 - 6H)t^3 + (H^2 + H/2)t^4 \\ &\quad + (-7H^2/2 + H/4)t^5 + (H^3 - H^2/2)t^6, \\ \beta_N &= (H - 3/2)t + 4Ht^2 - Ht^3 + 4H^2t^4 + (-H^3 + H^2/2)t^5. \end{aligned}$$

Uniformly for $1 \le H \le 4$, the remainders beyond the first terms have absolute values at most $(687/4)t^3$ and $156t^2$. Thus continuity gives a $\beta = 0$ source with $H \in [1, 2]$ and $D > 0$, and a $D = 0$ source with $H \in [2, 4]$ and $\beta > 0$. On the first face $p \le 7/8$ and $\lambda = 8DH + 2(H^2 - p)t \ge t/4$. On the second, $\beta\theta \ge t/16$, so $\lambda \ge 1/8 - 4t > 0$. All strict native gaps hold.

Finally take $\sigma = t$, $x = t^2/16$, $z = 1$, $b = 1/2$, $h = 3/4$. Then $H = 3/(4t)$ is unbounded along $t \downarrow 0$ and

$$D_N = \frac{-3t^2}{262144}(6t^5 - 61t^4 + 608t^3 - 7872t^2 + 8192t - 126976) > 0,$$

$$\beta_N = \frac{3(3t^2 + 64)(2t^3 + 5t^2 + 64)}{16384} > 0.$$

Also $K_A = O(\varepsilon)$ on this path by (6), so $c > 0$ is represented. These are singular limits inside the evaluated rectangle, not merely a nonsingular compact slice.

6 Exact checks and scope

The adjacent script `fixed_box_c1_checks.py` verifies with rational polynomial arithmetic the F_x norm 37, F_σ, the majorant norms 8590 and 5030, its testing support, all displayed rational margin comparisons, and the numerator formulas and sign brackets for the two faces. Its JSON output records each check. The inequalities above also supply the analytical proof; the script is not a numerical fit or a replacement for it.

This supplies the previously missing fixed constants and fixed testing radicals for the positive-angular main Gram. It does not compile the other retained square families, the $\lambda \leq 0$ branch, complementary testing charts, other C1 source charts or the global common-cover selection. None of those cases is silently included in the stated feature count.